

# Every Natural Strengthening Examined Is False: Type Mismatch in Lossless Reformulations of the Lonely Runner Conjecture

A sharp negative result with a mechanism: four settings across two reformulation worlds—the finite covering reformulation, the transfer-matrix cascade, character rigidity, and the Lonely Runner zonotope—in which every strengthening that would open a new toolkit and was examined is false, with exact deepest-hole values  $16/47 > 1/3$  and  $17/47 > 5/14$  and further exact witnesses from  $n = 5$  on, their denominators forced by a proved weight identity. With the program's bounded positive theorem and a certified deepest-hole archive.

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Reformulation Program (companion to the pair-sum monograph)  
A negative-results paper: no solution, no new cases of the conjecture; the surviving equivalent forms are as hard as they ever were.